

PA2: Wall-Following Robot

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Screen Recording: [Google Drive Link]

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Implementation

A ROS2 node implements right-hand wall following in Stage using a discrete PID controller, a five-state FSM, and adaptive velocity scaling. The robot subscribes to `/base_scan` and publishes `/cmd_vel` at 20 Hz.

PID Controller. Three LIDAR sectors extract d_{right} (-100° to -80°), d_{front} ($\pm 25^\circ$), and d_{fr} (-55° to -35°) as readings. The error $e(t) = d_{\text{target}} - d_{\text{right}}$ feeds a PID whose output is angular velocity ω . $\omega = K_p e + K_i \int e dt + K_d \dot{e}$, with $K_p=1.5$, $K_i=0.05$, $K_d=0.6$. Integral windup sets to ± 0.3 rad/s, and the derivative term is suppressed on the first tick after state transitions to avoid kick. A front avoidance bias adds up to $+0.8$ rad/s (left turn) as d_{front} drops below 0.8 m, allowing the PID to handle corners without FSM transitions. The d_{fr} sensor is used when below 0.8 m to prevent junction misreads from distant corridors.

FSM & Recovery. `FIND_WALL` rotates toward the closest LIDAR reading then drives forward; `ALIGN_LEFT` turns in place until the contacted wall is on the robot's right; `FOLLOW_WALL` runs the PID loop with adaptive velocity; `INTERNAL_CORNER` (triggered at $d_{\text{front}} < 0.35$ m) turns left, backing up when wedged ($d_{\text{front}} < 0.7$ m and $d_{\text{right}} < 0.5$ m); `SEARCH_WALL` (triggered at $d_{\text{right}} > 2.0$ m after 2 s dwell) turns right ($\omega = -0.5$ rad/s) for 4 s then drives toward the nearest wall. Minimum dwell times (0.5–2 s) on each state prevent rapid cycling at junctions, and 5–10 s timeouts on non-follow states ensure stuck recovery by falling back to `FIND_WALL`.

Adaptive Velocity. Linear velocity scales from $v_{\text{max}}=0.5$ m/s on straightaways to $v_{\text{min}}=0.05$ m/s at corners via a quadratic error curve: $v = v_{\text{max}} - \min(\frac{|e|}{0.5}, 1)^2 (v_{\text{max}} - v_{\text{min}})$, combined with a front-proximity slowdown that begins at 0.8 m.

Gain tuning. K_p was set first for adequate tracking response, K_d added to dampen oscillation, $\approx 0.4 K_p$, and K_i kept minimal to close steady-state offset. The quadratic velocity curve was chosen over linear because small PID error should not penalize large errors near corners.

Evaluation

Tests on `empty.world` and the `2017-02-11-00-31-57.world`.

World	Avg d_{right} (m)	Avg v (m/s)	Corner v (m/s)	FSM Transitions
Square room	0.50	0.49	0.14	0 (steady state)
Complex map (first lap)	0.57	0.42	0.08	~ 8
Complex map (multi-lap)	0.58	0.43	0.06	~ 25

In the square room, the PID tracks $d_{\text{target}}=0.50$ m with zero FSM chatter once settled—corners are handled by the front-avoidance bias within `FOLLOW_WALL`. On the complex map, the robot navigates corridors, internal corners, and external corners at T-junctions. Velocity reaches 0.49–0.50 m/s on straightaways and drops to 0.06–0.14 m/s at corners.

Credits

Developed by Xiaoming Zhao, referencing lecture materials, slides, and stage simulator documentation. Debugging, PID tuning, FSM design iteration, and report formatting assisted by Generative AI, including Claude and Gemini.